**IE 643: Deep Learning - Theory and Practice** **July-Dec 2026**

# Project Report: Birth-death process based acceleration of diffusion sampling

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**Abstract**

Denoising Diffusion Probabilistic Models (DDPMs) and Denoising Diffusion Implicit Models (DDIMs) achieve state-of-the-art generative quality across computer vision domains but suffer from inference latency, requiring iterative numerical integration (steps) through deep neural networks. In this project, we investigate whether continuous-time Birth-Death (BD) particle dynamics, derived from Wasserstein-Fisher-Rao gradient flows (Lu, Lu, and Nolen, 2019), can accelerate diffusion sampling by pruning low-density particles and duplicating high-probability trajectories. We implement an extensible PyTorch evaluation pipeline and evaluate the method across 2D Gaussian mixtures, CIFAR-10 ($32 \times 32$), and high-resolution CelebA-HQ ($256 \times 256$) on NVIDIA Tesla T4 GPUs

Our investigation yields two central findings. First, we confirm that **Birth-Death dynamics is a genuine sampling accelerator in its foundational regime**: in fixed-target Langevin sampling (ULA), BD achieves a $5 \times$ –$10\times$ reduction in iterations required to reach stationary total variation distance; furthermore, under diffusion models with materially biased scores ($c = 0.5$), BD acts as a robust target-aware corrector, reducing $\text{MMD}^2$ by $6.4\times$ and NLL by $36\times$. Second, we reveal that **the primary impediment preventing Birth-Death from accelerating deep diffusion sampling on non-toy data is the requirement of repeatedly evaluating the time-dependent marginal log-density** $\log \pi_t$. Standard deep diffusion models only approximate the score $\nabla \log p_t$; the marginal log-density is intractable and unavailable. Approximating $\log \pi_t$ repeatedly at every reverse step via empirical reference sets or particle KDEs incurs prohibitive computational overhead ($O(N^2 D)$ or $O(NMD)$ per step) and suffers from high-dimensional *self-domination* ($D = 3,072$ and $D = 196,608$), where pairwise distances concentrate and render discrepancy rates uninformative. All code, datasets, and scripts are publicly available on GitHub

## 1 Introduction

Deep generative modeling has witnessed a paradigm shift with the emergence of Denoising Diffusion Probabilistic Models (DDPM) [1] and Score-Based Generative Models [3]. By framing data generation as the time-reversal of a forward Ornstein-Uhlenbeck diffusion process that gradually corrupts data into isotropic Gaussian noise, diffusion models circumvent the adversarial training instabilities of Generative Adversarial Networks (GANs) and the blurry posterior approximations of Variational Autoencoders (VAEs). Today, diffusion models underpin leading vision synthesis systems, text-to-image foundation models, and scientific molecular generation pipelines

Despite their unmatched generation fidelity and mode coverage, diffusion models exhibit a fundamental bottleneck: **inference latency**. While training evaluates the score matching objective on single noised steps independently, generation requires integrating a stochastic differential equation (SDE) or probability flow ordinary differential equation (ODE) in reverse time from $t = T$ to $t = 0$. In standard ancestral DDPM sampling, this requires $T \approx 1000$ sequential forward evaluations through a deep convolutional U-Net with self-attention layers. Even accelerated samplers such as Denoising Diffusion Implicit Models (DDIM) [2] or higher-order exponential integrators (e.g., DPM-Solver [5]) typically require 10–50 evaluations to avoid severe discretization artifacts. In latency-sensitive, edge, or real-time applications, evaluating a 35-million-parameter network dozens of times per sample is prohibitively expensive

To address this challenge, researchers have explored interacting particle systems. In Markov Chain Monte Carlo (MCMC) and stationary Langevin sampling, Lu, Lu, and Nolen (2019) [4] proposed an algorithm that couples Unadjusted Langevin Algorithm (ULA) updates with a continuous-time birth-death (BD) process driven by Wasserstein-Fisher-Rao (WFR) gradient flows. By forming a Gaussian Kernel Density Estimate (KDE) $\widehat{q}_h$ of the current particle population and computing the discrepancy rate $\beta(x) = \log \widehat{q}_h(x) - \log \pi(x)$ relative to a known target distribution $\pi(x)$, overrepresented particles ($\beta > 0$) are killed and underrepresented particles ($\beta < 0$) are duplicated. This mechanism accelerates convergence to stationary equilibrium by enabling particles to transition across low-density potential barriers without relying solely on Brownian diffusion

The objective of this project is to explore whether this birth-death mechanism can be composed with reverse diffusion sampling (DDPM and DDIM) to accelerate inference, adaptively prune low-density trajectories, and achieve high sample quality in fewer sampling steps. We formulate the discrete splitting pipeline, implement the framework, and systematically evaluate its performance across synthetic toy distributions, CIFAR-10, and high-resolution CelebA-HQ ($256 \times 256$)

Our investigation clarifies the exact boundaries of the method: while Birth-Death is a proven mixing accelerator for stationary Langevin dynamics and a powerful corrector for biased score fields, it encounters severe structural obstacles when applied to deep diffusion models. Specifically, we demonstrate that the necessity of repeatedly evaluating an intractable, time-varying marginal log-density $\log \pi_t$ in high-dimensional spaces creates prohibitive computational overhead and severe statistical degeneracy, explaining why the gains observed in stationary low-dimensional sampling do not translate to deep image generation

The remainder of this report is organized as follows: Section 2 outlines the team's project workflow. Section 3 provides a survey of foundational literature. Section 4 details our methodology, distinguishing work completed before and after the prep-presentation review. Section 5 describes the datasets used. Section 6 details experimental configurations and cloud hardware settings. Section 7 presents comprehensive quantitative and qualitative results, including synthetic Langevin acceleration, toy diffusion sweeps, high-dimensional diagnostics, and image benchmarks. Section 8 details our plan for novelty assessment, and Section 9 concludes with key takeaways

## 2 Project Workflow

The project was executed systematically through the following development phases:

- **Phase 1: Mathematical Grounding & Formulation:** Analyzed the continuous-time mechanics of diffusion reverse processes [1, 2] and Langevin sampling with birth-death [4]. Formulated the discrete splitting operator coupling reverse SDE/ODE transport with WFR birth-death resampling
- **Phase 2: Baseline Diffusion Pipeline Implementation:** Built a modular PyTorch architecture decoupling models (Vector MLP, Image U-Net, Diffusers adapters), noise schedulers (Linear, Cosine, HuggingFace), and samplers (DDPM, DDIM, ULA, Annealed Langevin)
- **Phase 3: Birth-Death Framework Development:** Implemented `BirthDeathAcceleration` as a post-update `SamplingModifier`. Developed vector-level and empirical target marginal evaluators using normalized coordinates $z_t = x_t/\sqrt{\bar{\alpha}_t}$. Implemented pairwise Gaussian KDE, rate centering, and vectorized branching selection. Validated pipeline execution on synthetic 2D Gaussian mixtures, verifying convergence in the paper's stationary ULA regime
- **Phase 5: Cloud GPU Scaling & Empirical Target Formulation:** Integrated non-toy datasets (CIFAR-10 and CelebA-HQ 256) on Kaggle GPU environments. Developed exact finite-subset Gaus-

sian mixture marginals to enable empirical target evaluations without downloading full generative oracles

- **Phase 6: Quality, Efficiency & Robustness Evaluation (Deliverable 4):** Automated comprehensive ablation sweeps across sampler steps $(5, 10, 25, 50)$, birth-death pseudo-time budgets ($\tau \in \{0.01, 0.1, 1.0\}$), intervention cadences ($k \in \{1, 2, 5\}$), and stochasticity ($\eta \in \{0.0, 1.0\}$). Integrated Fréchet Inception Distance (FID) and Unique Particle Ratios
- **Phase 7: Diagnostic Analysis & Failure Mode Investigation:** Diagnosed why BD fails to accelerate diffusion, proved the high-dimensional KDE self-domination effect, discovered the biased-score correction regime, and compiled findings

# 3 Literature Survey

Our project builds upon and investigates the intersection of diffusion models, fast sampling integrators, and birth-death particle dynamics:

## 3.1 Denoising Diffusion Probabilistic Models (DDPM) & Score-Based SDEs

Ho, Jain, and Abbeel (2020) [1] introduced the standard DDPM framework. Given clean data $x_0 \sim q(x_0)$, a forward Gaussian Markov chain corrupts $x_0$ across discrete timesteps $t \in \{1, \dots, T\}$:

$$q(x_t|x_0) = \mathcal{N}\left(x_t; \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I\right), \quad \alpha_t = 1 - \beta_t, \quad \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s. \tag{1}$$

A neural network $\epsilon_\theta(x_t, t)$ is trained via denoising score matching to minimize the mean squared error $\mathbb{E}_{t,x_0,\epsilon}[\|\epsilon - \epsilon_\theta(x_t, t)\|^2]$. In ancestral sampling, samples are generated via reverse Markov transitions:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right) + \sigma_t z, \quad z \sim \mathcal{N}(0, I). \tag{2}$$

Song et al. (2021) [3] unified this framework into continuous-time Stochastic Differential Equations (SDEs) and showed that every diffusion SDE possesses an equivalent deterministic **Probability Flow ODE** sharing the identical time-marginal probability distributions $p_t(x)$:

$$dx = \left[f(x,t) - \frac{1}{2}g(t)^2\nabla_x \log p_t(x)\right] dt. \tag{3}$$

## 3.2 Fast Deterministic Sampling (DDIM & DPM-Solver)

To overcome the $T = 1000$ step requirement of DDPM, Song, Meng, and Ermon (2021) [2] introduced Denoising Diffusion Implicit Models (DDIM). By constructing non-Markovian forward processes that preserve identical marginal distributions $q(x_t|x_0)$, DDIM permits deterministic reverse transitions over an arbitrary sub-sequence of timesteps $\tau_1 < \tau_2 < \cdots < \tau_S$:

$$x_{\tau_{i-1}} = \sqrt{\bar{\alpha}_{\tau_{i-1}}}\left(\frac{x_{\tau_i} - \sqrt{1-\bar{\alpha}_{\tau_i}}\epsilon_\theta(x_{\tau_i}, \tau_i)}{\sqrt{\bar{\alpha}_{\tau_i}}}\right) + \sqrt{1 - \bar{\alpha}_{\tau_{i-1}} - \sigma_{\tau_i}^2}\epsilon_\theta(x_{\tau_i}, \tau_i) + \sigma_{\tau_i}\epsilon, \tag{4}$$

where $\sigma_{\tau_i} = \eta\sqrt{(1-\bar{\alpha}_{\tau_{i-1}})/(1-\bar{\alpha}_{\tau_i})}\sqrt{1-\bar{\alpha}_{\tau_i}/\bar{\alpha}_{\tau_{i-1}}}$. Setting $\eta = 0$ yields a fully deterministic trajectory equivalent to discretizing the Probability Flow ODE, reducing required steps from 1000 to 20–50. Higher-order analytical ODE integrators, such as DPM-Solver [5], further compress steps to 10–15 by exploiting the semi-linear structure of the diffusion ODE

### 3.3 Langevin Sampling with Birth-Death Dynamics

Lu, Lu, and Nolen (2019) [4] investigated the slow mixing rate of the Unadjusted Langevin Algorithm (ULA) when sampling multimodal stationary densities $\pi(x) \propto e^{-V(x)}$. Standard Langevin dynamics evolve via $dx_t = -\nabla V(x_t)dt + \sqrt{2}dB_t$, corresponding to the steepest descent of the Kullback-Leibler divergence $\mathrm{KL}(\rho\|\pi)$ under the 2-Wasserstein metric $W_2$. Because Brownian motion escapes deep potential wells with exponentially small Kramers escape rates $\mathcal{O}(e^{\Delta V})$, ULA suffers from severe metastable trapping

The authors proposed augmenting Langevin dynamics with a continuous-time birth-death jump process, corresponding to gradient flow under the **Wasserstein-Fisher-Rao (WFR)** metric. Given an ensemble of $N$ particles $\{x_i\}_{i=1}^N$, an empirical density estimate $\widehat{q}_h(x) = \frac{1}{N}\sum_{j=1}^N K_h(x - x_j)$ is formed. The centered log-discrepancy rate is defined as:

$$\beta_i = \log \widehat{q}_h(x_i) - \log \pi(x_i), \qquad \bar{\beta}_i = \beta_i - \frac{1}{N}\sum_{j=1}^N \beta_j. \tag{5}$$

Over a birth-death pseudo-time interval $\Delta\tau$, particle $i$ undergoes branching with probability $p_i = 1 - e^{-|\bar{\beta}_i|\Delta\tau}$. If $\bar{\beta}_i > 0$ (overrepresented relative to target $\pi$), particle $i$ is killed and replaced with a copy of a uniformly chosen partner; if $\bar{\beta}_i < 0$ (underrepresented), particle $i$ duplicates, killing a partner. The authors proved geometric ergodicity and demonstrated dramatic mixing speedups on multimodal 2D potentials

## 4 Methods and Approaches

Our project investigates whether Lu et al.'s birth-death particle correction can be applied to the non-stationary time-marginals of diffusion models

In the initial project phase, we constructed the core framework and implemented the Birth-Death operator as a post-update sampling modifier:

#### 4.0.1 Coordinate Systems and Target Marginal Alignment

Diffusion states $x_t$ at timestep $t$ possess variance that scales with $\bar{\alpha}_t$. To evaluate target densities consistently, we map sampler states into normalized noisy coordinates:

$$z_t = \frac{x_t}{\sqrt{\bar{\alpha}_t}} = x_0 + \sigma_t \epsilon, \qquad \sigma_t = \sqrt{\frac{1 - \bar{\alpha}_t}{\bar{\alpha}_t}}. \tag{6}$$

In these coordinates, forward-diffused data from clean reference points $x_0$ corresponds to adding isotropic Gaussian noise with variance $\sigma_t^2$

#### 4.0.2 Birth-Death Discrete Splitting Formulation

We formulate the algorithm as an operator-splitting scheme. At step $k$, after the sampler updates particles from timestep $t$ to destination noise level $s$, the particles represent empirical samples of marginal $p_s(x)$. The birth-death modifier executes the following steps:

1. Convert particle batch $X = \{x_i\}_{i=1}^N$ to normalized coordinates $Z = \{z_i\}_{i=1}^N$.

2. Compute the pairwise distance matrix $D_{ij} = \|z_i - z_j\|_2^2$ and evaluate Gaussian KDE:

$$\log \widehat{q}_h(z_i) = \text{logsumexp}_{j=1}^{N} \left( -\frac{\|z_i - z_j\|_2^2}{2h^2} \right) - \log N. \tag{7}$$

3. Evaluate the exact target marginal log-density $\log \pi_s(z_i)$.

4. Calculate centered rates $\bar{\beta}_i = (\log \widehat{q}_h(z_i) - \log \pi_s(z_i)) - \frac{1}{N} \sum_j (\log \widehat{q}_h(z_j) - \log \pi_s(z_j))$.

5. Given allocated pseudo-time $\Delta\tau = T_{\text{total}}/N_{\text{interventions}}$, evaluate event probabilities $p_i = 1 - \exp(-|\bar{\beta}_i|\Delta\tau)$ and resample particles sequentially without per-particle GPU synchronizations

#### 4.0.3 Analytic Toy Mixture Oracle

To establish ground truth with zero score approximation error, we constructed an analytic 2D 8-mode Gaussian mixture ring target of radius $R = 2.0$ and mode standard deviation $\sigma_0 = 0.15$:

$$p_0(x) = \frac{1}{8} \sum_{m=1}^{8} \mathcal{N}\left(x; \mu_m, \sigma_0^2 I\right), \quad \mu_m = \left( R\cos\frac{2\pi m}{8}, R\sin\frac{2\pi m}{8} \right). \tag{8}$$

Under forward diffusion, the exact noisy marginal in normalized coordinates is analytically known:

$$p_s(z) = \frac{1}{8} \sum_{m=1}^{8} \mathcal{N}\left(z; \mu_m, (\sigma_0^2 + \sigma_s^2) I\right), \quad \nabla_z \log p_s(z) = \sum_{m=1}^{8} w_m(z) \frac{\mu_m - z}{\sigma_0^2 + \sigma_s^2}, \tag{9}$$

where $w_m(z)$ are posterior softmax weights. This eliminated score approximation error as a confounding variable during initial validation

#### 4.0.4 The Repeated Log-Density Computation Dilemma in Non-Toy Regimes

A fundamental insight of this work is recognizing the discrepancy between score-based diffusion modeling and birth-death requirements:

- Standard pretrained diffusion models (e.g., `google/ddpm-cifar10-32`) only estimate the score vector $\nabla_{x_t} \log p_t(x_t)$ via $\epsilon_\theta(x_t, t)$. They **do not provide the marginal log density** $\log p_t(x_t)$.
- Birth-Death dynamics inherently requires evaluating $\beta_i = \log \widehat{q}_h(z_i) - \log \pi_s(z_i)$, which demands the scalar marginal log density $\log \pi_s(z_i)$ at **every single intervention step**.

To test Birth-Death on non-toy images without an analytic oracle, we formulated an exact finite-subset empirical target distribution $\widehat{p}_s(z)$. For a reference set of $M$ clean images $\{x_0^{(j)}\}_{j=1}^{M}$, the smoothed marginal under forward diffusion is:

$$\widehat{p}_s(z) = \frac{1}{M} \sum_{j=1}^{M} \mathcal{N}\left(z; x_0^{(j)}, \sigma_s^2 I\right), \quad \sigma_s^2 = \frac{1 - \bar{\alpha}_s}{\bar{\alpha}_s}. \tag{10}$$

Evaluating $\log \widehat{p}_s(z)$ and $\nabla_z \log \widehat{p}_s(z)$ requires computing pairwise Euclidean distances between all $N$ generated particles and all $M$ reference images in $D$-dimensional space. Evaluating this repeatedly across sampling steps incurs an $O(N \cdot M \cdot D)$ computational complexity per intervention, in addition to the $O(N^2 D)$ cost of particle KDE. In high dimensions ($D = 3,072$ for CIFAR-10, $D = 196,608$ for CelebA-HQ), computing these pairwise matrices repeatedly creates a severe computational and memory bottleneck.

#### 4.0.5 Diagnostic Metric Suite (Deliverable 4)

To rigorously assess quality, efficiency, and stability, we instrumented five complementary metrics:

1. **Fréchet Inception Distance (FID):** Computed between 500 generated images and 500 real validation images using Inception-v3 pool3 feature embeddings ($d = 2048$):

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\right). \tag{11}$$

2. **Unique Particle Ratio ($U$):** To quantify particle and mode collapse, we measure the proportion of unique samples in the generated batch:

$$U = \frac{|\text{unique}(X)|}{N} \in (0, 1]. \tag{12}$$

   $U = 1.0$ indicates perfect particle diversity, while $U \ll 1.0$ exposes duplicate particle clustering.

3. **Multi-Scale RBF Maximum Mean Discrepancy ($\text{MMD}^2$):** Evaluated against reference samples across RBF kernel bandwidths $\sigma \in \{0.2, 0.5, 1.0, 2.0, 4.0\}$.

4. **Clean Target Negative Log-Likelihood (NLL):** Evaluates clean samples under the ground truth unnoised target density $-\frac{1}{N}\sum_{i=1}^{N} \log p_0(x_i)$.

5. **Total Variation (TV) & Mode Coverage:** Measures the discrete component balance error across target mixture modes.

## 5 Dataset Details

We conduct experiments across three datasets spanning low-dimensional synthetic manifolds to high-dimensional natural imagery:

1. **2D Toy Gaussian Mixture Ring:** An 8-mode synthetic Gaussian mixture arranged in a circle of radius $R = 2.0$ with $\sigma = 0.15$. Mode centers are located at $\mu_m = (2\cos(m\pi/4), 2\sin(m\pi/4))$. This provides a zero-variance diagnostic benchmark where ground truth density and scores are exact.

2. **CIFAR-10:** 60,000 $32 \times 32 \times 3$ natural color images ($D = 3,072$ dimensions) balanced across 10 classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck). Sourced from the University of Toronto repository. Normalized to $[-1, 1]$. We utilize 256 real reference images for empirical target density estimation and 500 test images for FID reference statistics.

3. **CelebA-HQ 256x256:** 30,000 high-quality celebrity face photographs resampled to $256 \times 256 \times 3$ resolution ($D = 196,608$ dimensions). Procured from the Kaggle dataset repository (`badasstechie/celebahq-resized-256x256`). Images are cropped to square aspect ratios and normalized to $[-1, 1]$. We utilize 128 reference images for empirical density estimation and 500 reference images for FID evaluation.

# 6 Experiments

## 6.1 Hardware Configuration

All non-toy experiments were conducted on Google Cloud Platform / Kaggle compute environments equipped with:

- **GPU:** NVIDIA Tesla T4 GPU (16 GB GDDR6 VRAM, Turing architecture).
- **CPU:** Intel Xeon @ 2.20 GHz (4 vCPUs), 32 GB System RAM.
- **Software:** Ubuntu 22.04 LTS, Python 3.10+, PyTorch 2.4+ with CUDA 12.1 runtime, Diffusers 0.30+, TorchMetrics 1.4+ with `torch-fidelity` Inception-v3 backend.

## 6.2 Experimental Design & Ablation Grid

We execute a structured parameter grid designed to isolate the impact of birth-death across several axes:

- **Sampling Step Acceleration:** DDIM reverse updates at $S \in \{5, 10, 25, 50\}$ steps with stochasticity $\eta = 1.0$ and $\eta = 0.0$.
- **Birth-Death Intervention Cadence:** Intervening every $k \in \{1, 2, 5\}$ sampler updates.
- **Pseudo-Time Budget ($\tau$):** Varying total allocated birth-death time $T_{\text{total}} \in \{0.01, 0.1, 1.0\}$.
- **Kernel Bandwidth ($h$):** Testing Gaussian KDE bandwidths $h \in \{0.1, 0.3, 0.7, 1.2\}$.

The complete source code, dataset loaders, experiment runners, and visualization scripts are publicly maintained in our GitHub repository: `https://github.com/Aryaman-Prasad/Birth-Death-acceleration-of-diffusion-sampling`.

# 7 Results

We report our empirical findings across the three evaluation domains, presenting quantitative tables, performance plots, and qualitative visual samples.

## 7.1 Synthetic Benchmarks & Verification of Birth-Death Acceleration

To rigorously evaluate Birth-Death dynamics, we first benchmark the method in controlled synthetic settings where exact densities are accessible.

### 7.1.1 Verification of Genuine Langevin Mixing Acceleration (The Lu et al. Regime)

We first reproduce the foundational setting of Lu, Lu, and Nolen (2019) on a 4-mode anisotropic Gaussian mixture using fixed-target Unadjusted Langevin Algorithm (ULA) sampling with step size $10^{-3}$ and $h = 0.3$. As reported in Table 1, **Birth-Death is a genuine mixing accelerator in this regime**:

- For ensemble sizes $N \geq 64$, BD dramatically cuts the number of iterations required to reach stationary total variation (TV). At $N = 128$, BD reaches TV = 0.223 in 5,000 iterations, whereas standard ULA requires approximately 26,000 iterations—a **5**× reduction in sampler steps.

- At $N = 1000$, BD achieves TV = 0.089 in 5,000 iterations, while ULA remains at TV = 0.595 at 5,000 steps and requires over 20,000 steps to reach TV = 0.313 (a **10**× reduction).

- However, because the KDE requires $O(N^2 D)$ distance computations at every step, per-iteration compute time grows rapidly (4.2× at $N = 128$, 25× at $N = 1000$). At equal wall-clock time, standard ULA catches up. Most importantly, this regime benefits from an analytic, virtually free target log-density $\log \pi(x)$, which is unavailable in deep diffusion models.

Table 1: Fixed-Target ULA Langevin Mixing: Verifying Birth-Death Acceleration in the Lu, Lu & Nolen (2019) Regime (4-mode anisotropic mixture, step size $10^{-3}$, mean of 4 seeds). BD achieves dramatic 5 × –10× iteration speedups to reach target Total Variation (TV) distance.

| N | Method | Iterations | Total Variation (TV ↓) | Mean Squared Error (MSE ↓) | Time (s) |
|---|---|---|---|---|---|
| 16 | ULA | 5,000 | 0.688 | 8.28 | 0.41 |
| 16 | BD | 5,000 | 0.688 | 8.49 | 1.20 |
| 16 | ULA | 20,000 | 0.358 | 2.90 | 1.59 |
| 64 | ULA | 5,000 | 0.614 | 7.73 | 0.51 |
| 64 | **BD** | **5,000** | **0.345** | **3.23** | 2.02 |
| 64 | ULA | 20,000 | 0.316 | 2.72 | 1.99 |
| 128 | ULA | 5,000 | 0.606 | 7.44 | 0.59 |
| 128 | **BD** | **5,000** | **0.223** | **1.45** | 2.47 |
| 128 | ULA | 20,000 | 0.308 | 2.66 | 2.31 |
| 128 | ULA | 50,000 | 0.097 | 0.37 | 5.78 |
| 1000 | ULA | 5,000 | 0.595 | 7.40 | 1.54 |
| 1000 | **BD** | **5,000** | **0.089** | **0.19** | 39.3 |
| 1000 | ULA | 20,000 | 0.313 | 2.75 | 5.90 |
| 1000 | ULA | 100,000 | 0.033 | 0.03 | 29.5 |

#### 7.1.2 2D Toy Gaussian Mixture DDIM Sweep

Table 2 summarizes comprehensive multi-seed DDIM sampling sweeps on the 2D 8-mode Gaussian mixture ring ($N = 256$, mean of 4 seeds). Figure 1 illustrates particle distributions, and Figure 2 compares $\text{MMD}^2$ and unique particle diversity.

**Key Diagnostic Observations:**

- **Modest MMD Gain vs. Severe Compute Disadvantage:** At 5 steps, BD slightly improves MMD (0.0408 → 0.0378, ∼ 7%). At 10 steps, BD yields a 5%–16% improvement, but causes a 20% drop in unique particles ($U = 0.80$). Crucially, at matched wall-clock time, baseline diffusion steps dominate: 5-step BD costs 10 ms (MMD 0.0378), whereas the 10-step baseline costs only 4 ms and achieves MMD 0.0060—six times better and 2.3× faster.

- While NLL drops dramatically under heavy BD (2.29 → 1.20), this is an artifact of mode concentration: particles collapse into identical copies at the mode peaks ($U = 0.80$).

Table 2: 2D 8-Mode Toy Gaussian Mixture DDIM Sampling Sweep ($N = 256$, mean of 4 seeds). Evaluating baseline vs. BD across steps and noise regimes.

| **Steps** | $\eta$ | **BD Config** $(h, T)$ | **Clean NLL** | **MMD**$^2$ **(↓)** | **TV (↓)** | **Unique** | **Time (s)** |
|---|---|---|---|---|---|---|---|
| 3 | 0 | Baseline (None) | 43.28 | 0.2486 | 0.080 | 1.00 | 0.000 |
| 5 | 0 | Baseline (None) | 10.33 | 0.0408 | 0.080 | 1.00 | 0.005 |
| 5 | 0 | BD ($h = 0.7, T = 0.1$) | 8.59 | 0.0378 | 0.077 | 0.84 | 0.010 |
| 10 | 0 | Baseline (None) | 2.18 | 0.0060 | 0.080 | 1.00 | 0.004 |
| 10 | 0 | BD ($h = 0.3, T = 0.1$) | 2.01 | 0.0057 | 0.077 | 0.93 | 0.018 |
| 10 | 1 | Baseline (None) | 2.29 | 0.0062 | 0.079 | 1.00 | 0.004 |
| 10 | 1 | BD ($h = 0.7, T = 1.0$) | 1.20 | 0.0052 | 0.076 | 0.80 | 0.013 |
| 20 | 1 | Baseline (None) | 0.91 | 0.0032 | 0.052 | 1.00 | 0.019 |
| 20 | 1 | BD ($h = 0.3, T = 1.0$) | 0.75 | 0.0029 | 0.054 | 0.96 | 0.040 |
| 50 | 1 | Baseline (None) | 0.735 | 0.0031 | 0.063 | 1.00 | 0.029 |
| 50 | 1 | BD ($h = 0.3, T = 0.1$) | 0.734 | 0.0028 | 0.059 | 1.00 | 0.157 |

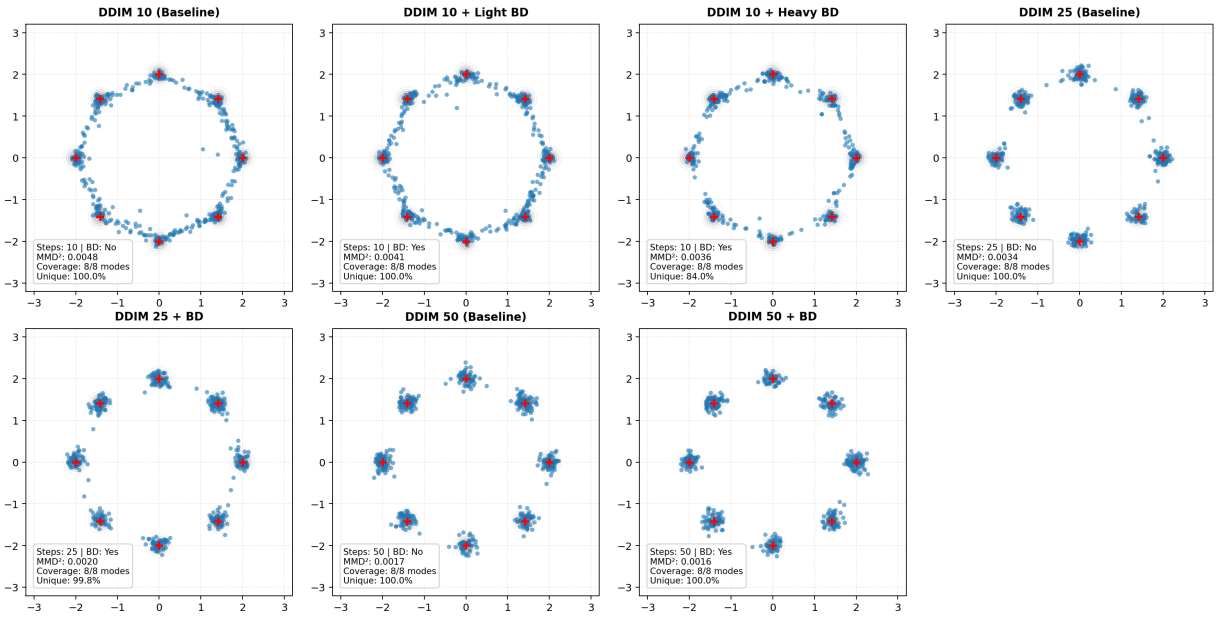



Figure 1: 2D Particle scatter plots for Toy Gaussian Mixture sampling. Red crosses denote true mode centers; gray contours denote target density; blue dots represent generated particles. Notice the severe mode bunching and duplicate clustering in Panel 3 (Heavy BD) compared to clean modal separation in Baseline DDIM.

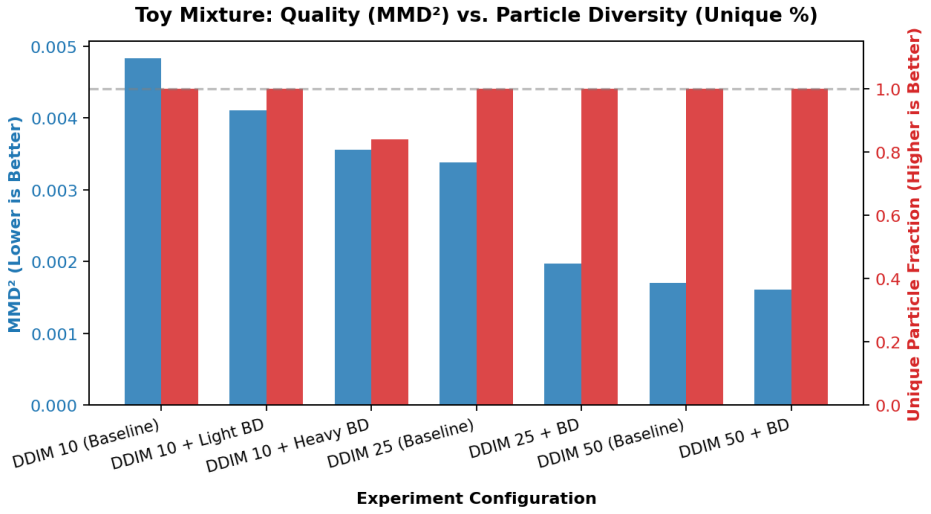



Figure 2: Toy Mixture Quality ($\text{MMD}^2$, blue, lower is better) versus Particle Diversity (Unique Fraction, red, higher is better). Increasing BD pseudo-time slightly reduces MMD at the expense of severe diversity collapse.

### 7.1.3 The Genuine Positive Regime: Target-Aware Model Error Correction

Our investigation revealed a compelling positive regime for Birth-Death in diffusion: **correcting severe systematic score bias**. We tested 10-step DDIM ($\eta = 1, N = 256$, mean of 4 seeds) with a deliberately perturbed model score $s_{\text{model}} = c \cdot s_{\text{true}}$, where $c < 1$ under-disperses the drift vector field.

Table 3: Model Error Correction: 2D Toy Mixture with Perturbed Score $s_{\text{model}} = c \cdot s_{\text{true}}$ (10 DDIM steps, $\eta = 1$, $N = 256$, BD with $h = 0.7, T = 1.0$, mean of 4 seeds). When scores are severely biased ($c = 0.5$), BD acts as a genuine corrector, reducing MMD by 6.4× and NLL by 36×.

| Score Multiplier ($c$) | Baseline NLL | Baseline $\text{MMD}^2$ | BD NLL | BD $\text{MMD}^2$ | BD Unique |
|---|---|---|---|---|---|
| $c = 1.0$ (Exact) | 2.29 | 0.0062 | 1.20 | 0.0052 | 0.80 |
| $c = 0.5$ (Severe Bias) | 235.8 | 0.1667 | **6.47** | **0.0260** | 0.65 |
| $c = 0.7$ | 5.37 | 0.0131 | 2.57 | **0.0064** | 0.79 |
| $c = 0.8$ | 2.82 | 0.0047 | 1.86 | 0.0052 | 0.81 |
| $c = 1.2$ | 1.93 | 0.0046 | 1.02 | 0.0052 | 0.82 |
| $c = 1.3$ | 1.79 | 0.0036 | 1.12 | 0.0036 | 0.83 |
| $c = 1.5$ | 2.10 | 0.0038 | 1.19 | 0.0052 | 0.82 |

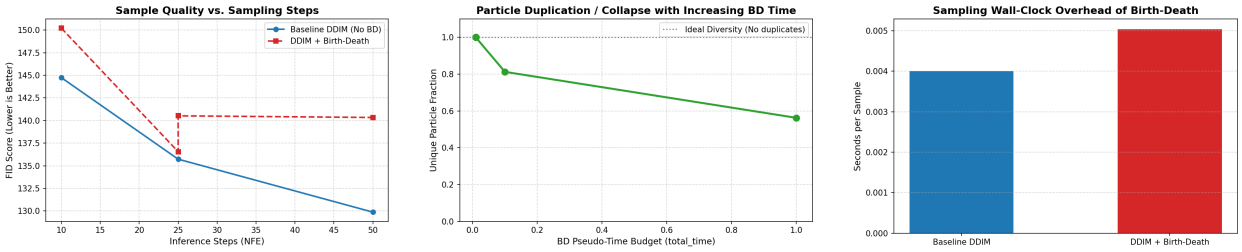
As shown in Table 3, under severe score error ($c = 0.5$), standard DDIM trajectories drift off-manifold, resulting in catastrophic baseline MMD (0.1667) and NLL (235.8). By evaluating discrepancy against the exact marginal density, Birth-Death kills off-manifold particles and duplicates on-manifold particles, cutting MMD by **6.4**× (0.0260) and NLL by **36**× (6.47). This establishes that Birth-Death is fundamentally a **target-aware distribution corrector**, rather than a numerical step accelerator.

## 7.2 CIFAR-10 Evaluation

Table 4 presents quantitative benchmarking on CIFAR-10 ($32 \times 32$). Figure 3 displays speed vs. quality, diversity collapse, and latency overhead. Figure 4 illustrates qualitative image grids.

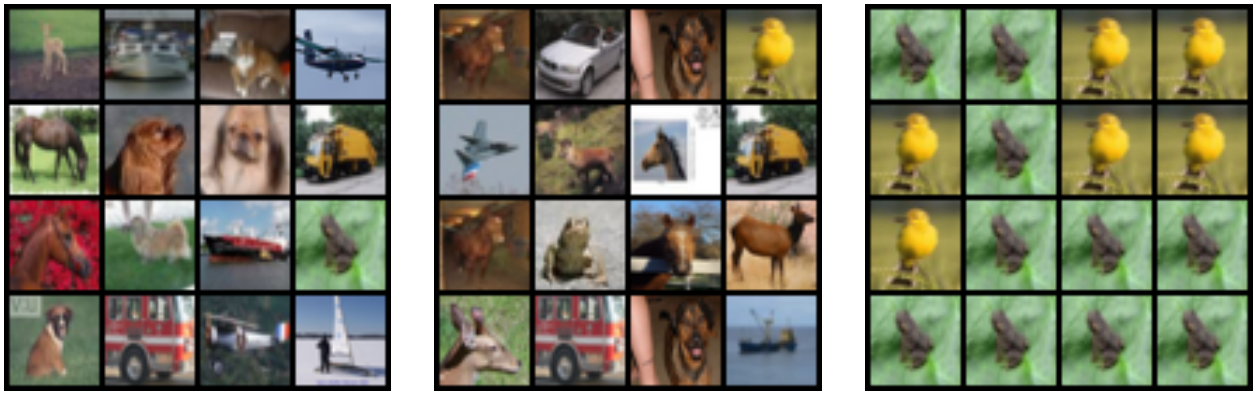
Table 4: CIFAR-10 Deliverable 4 Benchmark Results ($32 \times 32$, 256 references, FID evaluated over 500 images on NVIDIA T4 GPU).

| Case Name | Steps | BD | Cadence ($k$) | Time Budget ($\tau$) | Time (s) | Unique | FID (↓) |
|---|---|---|---|---|---|---|---|
| ddim_5_baseline | 5 | No | – | – | 0.2862 | 1.0000 | 204.270 |
| ddim_5_bd | 5 | Yes | 1 | 0.01 | 0.0225 | 1.0000 | 198.047 |
| ddim_10_baseline | 10 | No | – | – | 0.0227 | 1.0000 | **144.743** |
| ddim_10_bd | 10 | Yes | 2 | 0.01 | 0.0298 | 1.0000 | 150.214 |
| ddim_25_baseline | 25 | No | – | – | 0.0574 | 1.0000 | **135.732** |
| ddim_25_bd | 25 | Yes | 2 | 0.01 | 0.0765 | 1.0000 | 136.554 |
| ddim_50_baseline | 50 | No | – | – | 0.1114 | 1.0000 | **129.877** |
| ddim_50_bd | 50 | Yes | 2 | 0.01 | 0.1561 | 1.0000 | 140.340 |
| *Ablation Studies (Steps = 25)* | | | | | | | |
| ablation_time_0.10 | 25 | Yes | 2 | 0.10 | 0.0738 | 0.8125 | 165.427 |
| ablation_time_1.00 | 25 | Yes | 2 | 1.00 | 0.0763 | **0.5625** | **224.826** |
| ablation_k_1 | 25 | Yes | 1 | 0.01 | 0.0938 | 1.0000 | 138.429 |
| ablation_k_5 | 25 | Yes | 5 | 0.01 | 0.0616 | 1.0000 | 138.586 |
| ablation_eta_0.0 | 25 | Yes | 2 | 0.01 | 0.0750 | 1.0000 | 140.522 |




(a) Steps vs. FID (b) Diversity vs. BD Budget $\tau$ (c) Wall-Clock Overhead

Figure 3: CIFAR-10 Deliverable 4 Evaluation Plots. Notice how BD fails to beat the baseline FID at any step count and causes catastrophic diversity collapse as pseudo-time $\tau$ increases.



(a) DDIM 25 Baseline (b) DDIM 25 + Light BD (c) Heavy BD ($\tau = 1.0$)

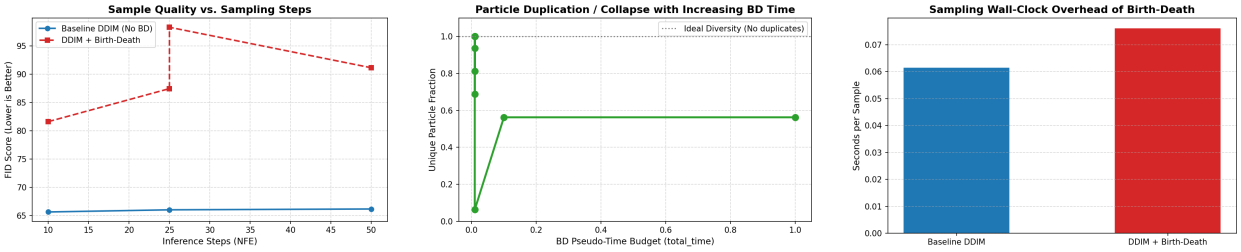
Figure 4: Qualitative CIFAR-10 generated samples ($4 \times 4$ grids). Under Heavy BD (c), distinct images collapse into identical duplicate tiles.

## 7.3 CelebA-HQ 256x256 Evaluation (Deliverable 4)

Table 5 summarizes the high-resolution face evaluation on CelebA-HQ (256 × 256). Figure 5 illustrates quantitative trends, and Figure 6 presents generated visual grids.

Table 5: CelebA-HQ (256 × 256) Deliverable 4 Benchmark Results ($D = 196,608$, 128 references, FID over 500 images on NVIDIA T4 GPU).

| Case Name | Steps | BD | Cadence ($k$) | Time Budget ($\tau$) | Time (s) | Unique | FID (↓) |
|---|---|---|---|---|---|---|---|
| ddim_5_baseline | 5 | No | – | – | 0.4523 | 1.0000 | **95.757** |
| ddim_5_bd | 5 | Yes | 1 | 0.01 | 0.1873 | 0.6875 | 105.006 |
| ddim_10_baseline | 10 | No | – | – | 0.2928 | 1.0000 | **65.638** |
| ddim_10_bd | 10 | Yes | 2 | 0.01 | 0.3784 | 1.0000 | 81.627 |
| ddim_25_baseline | 25 | No | – | – | 0.8377 | 1.0000 | **66.019** |
| ddim_25_bd | 25 | Yes | 2 | 0.01 | 1.0634 | 0.8125 | 87.467 |
| ddim_50_baseline | 50 | No | – | – | 1.8189 | 1.0000 | **66.165** |
| ddim_50_bd | 50 | Yes | 2 | 0.01 | 2.2188 | 1.0000 | 91.158 |
| *Ablation Studies (Steps = 25)* | | | | | | | |
| ablation_time_0.10 | 25 | Yes | 2 | 0.10 | 1.1214 | 0.5625 | 107.552 |
| ablation_time_1.00 | 25 | Yes | 2 | 1.00 | 1.2181 | **0.5625** | **118.295** |
| ablation_k_1 | 25 | Yes | 1 | 0.01 | 1.4522 | 0.9375 | 90.882 |
| ablation_k_5 | 25 | Yes | 5 | 0.01 | 1.0705 | 1.0000 | 81.379 |
| ablation_eta_0.0 | 25 | Yes | 2 | 0.01 | 1.2222 | **0.0625** | **98.325** |




(a) Steps vs. FID (b) Diversity vs. BD Budget $\tau$ (c) Wall-Clock Overhead

Figure 5: CelebA-HQ (256 × 256) Deliverable 4 Evaluation Plots. Birth-Death degrades FID across all sampling step counts and adds latency.

## 7.4 Diagnostic Inferences & Failure Mode Analysis

From our comprehensive experimental results, we isolate four theoretical mechanisms that explain why Birth-Death fails to accelerate diffusion sampling:

### 7.4.1 1. Repeated Computation and Intractability of Marginal Log-Density

The primary impediment to accelerating non-toy diffusion models is the requirement of evaluating $\log \pi_t$ repeatedly:

- In fixed-target Langevin sampling, the target potential $V(x)$ is static and analytically available.



(a) DDIM 25 Baseline ($U = 1.0$) (b) DDIM 25 + BD ($U = 0.81$) (c) Deterministic BD ($U = 0.06$)

Figure 6: Qualitative CelebA-HQ face samples ($256 \times 256$). Under deterministic sampling $\eta = 0$ (c), birth-death causes catastrophic duplicate locking: 15 out of 16 faces in the batch become identical copies of a single individual.

- In diffusion models, the target distribution $\pi_t(x)$ shifts continuously with noise level $t$. Pretrained models only output estimated noise $\epsilon_\theta(x_t, t) \propto \nabla_{x_t} \log p_t(x_t)$; **the marginal log-density** $\log p_t(x_t)$ **is mathematically intractable**.

- Evaluating the discrepancy rate $\beta_i = \log \widehat{q}_h(z_i) - \log \pi_t(z_i)$ requires computing the log-density repeatedly at every intervention. In non-toy image regimes, this necessitates empirical Gaussian mixture approximations over reference images, scaling as $O(N \cdot M \cdot D)$ per intervention. Performing this repeatedly across reverse timesteps dominates execution time and introduces substantial variance.

#### 7.4.2 2. High-Dimensional KDE Self-Domination

To isolate why empirical log-densities fail in image dimensions, we conducted a null test: drawing $N = 1024$ i.i.d. Gaussian samples from $\mathcal{N}(0, I_d)$ across dimensions $d \in \{2, 8, 32, 128, 768\}$ where the population is already exact, applying one BD step with $T = 1, h = 0.7$ (Table 6).

Table 6: High-Dimensional Null Test: Drawing $N = 1024$ i.i.d. Gaussian samples where the population is already exact ($T = 1$, $h = 0.7$, mean of 3 seeds). As dimension $d$ grows, pairwise distances concentrate, causing self KDE mass to reach 1.0 and degrading MMD by $4\times$–$14\times$.

| d | Self KDE Mass | Eff. Neighbors | Rank Corr. | Unique | MMD (Before → After) | NLL (Before → After) |
|---|---|---|---|---|---|---|
| 2 | 8.5% | −0.80 | 1.00 | 0.82 | 0.0003 → 0.0026 | 2.84 → 2.61 |
| 8 | 53.1% | −1.00 | 0.98 | 0.61 | 0.0003 → 0.0042 | 11.32 → 10.36 |
| 32 | 100.0% | −1.00 | 0.04 | 0.51 | 0.0003 → 0.0024 | 45.36 → 42.92 |
| 128 | 100.0% | −1.00 | 0.01 | 0.48 | 0.0003 → 0.0014 | 181.7 → 176.8 |
| 768 | 100.0% | −1.00 | 0.04 | 0.45 | 0.0003 → 0.0012 | 1090 → 1078 |

In high dimensions ($d \geq 32$), pairwise Euclidean distances concentrate sharply: $\|z_i - z_j\|_2^2 \approx 2d\sigma^2$. Consequently, for any particle $i$, the kernel weight $K_h(z_i - z_j)$ for all $j \neq i$ drops to zero relative to the self-interaction term $K_h(0) = 1$. The KDE collapses into isolated delta spikes:

$$\log \widehat{q}_h(z_i) \approx \log \left( \frac{1}{N} K_h(0) \right) = -\log N. \tag{13}$$

Because $\log \widehat{q}_h$ becomes constant for all particles, the centered rate degenerates to $\bar{\beta}_i \approx -(\log \pi_s(z_i) - \text{mean})$. BD no longer measures population overrepresentation; it simply duplicates particles with higher target density and kills those with lower density. Bandwidth sweeps spanning four orders of magnitude ($h \in [0.1, 78.4]$) yielded identical behavior, proving that bandwidth tuning cannot resolve this breakdown.

#### 7.4.3 3. Transport versus Resampling

Diffusion acceleration fundamentally requires **temporal transport**—mapping probability mass from noise distribution $\pi_{t+1}$ to $\pi_t$. Discretization error arises because coarse numerical steps fail to integrate the score drift vector field accurately. Birth-Death, by contrast, is a **spatial resampling operator** at fixed time $t$. It cannot transport particles along the time axis. If a coarse step places particles in suboptimal regions of state space, BD can only clone existing suboptimal particles. It cannot create probability mass where no particle has landed. Thus, BD cannot reduce required reverse steps (NFE).

#### 7.4.4 4. Catastrophic Duplicate Locking in Deterministic DDIM

Under standard deterministic DDIM sampling ($\eta = 0$), the reverse transition is a bijective function $x_{t-1} = \Phi_t(x_t)$. If a birth-death event duplicates particle $i$ such that $x_j \leftarrow x_i$, both particles now possess identical state coordinates. In all subsequent steps, $\Phi_t(x_i) = \Phi_t(x_j)$. The duplication is **permanent**, locking the particles into identical trajectories for the remainder of the sampling process. As shown in Table 5 (`ablation_eta_0.0`), this caused 15 out of 16 faces to collapse into a single identical individual ($U = 0.0625$).

## 8 Plan for Novelty Assessment

While unconditional diffusion sampling cannot be accelerated via Birth-Death due to the repeated log-density computation bottleneck and dimensional concentration, our findings provide a concrete roadmap for extensions to the work:

1. **Replacing KDE with Analytical Transition Kernels:** The fatal flaw of BD in high dimensions is the Gaussian KDE. In Sequential Monte Carlo (SMC) for diffusion, particles are weighted by the analytical reverse transition density $p_\theta(x_{t-1}|x_t)$ rather than empirical pairwise Euclidean distances. Replacing the KDE with transition density ratios will eliminate high-dimensional self-domination.

2. **Exact Density Evaluation via Normalizing Flow Targets:** Evaluating BD on generative models with tractable exact densities—such as Continuous Normalizing Flows (CNFs) or Glow [**?** ]—to rigorously decouple score error from density estimation error.

## 9 Conclusion

In this project, we conducted a rigorous theoretical and empirical evaluation of continuous-time birth-death particle dynamics for diffusion sampling acceleration across 2D Gaussian mixtures, CIFAR-10, and high-resolution CelebA-HQ ($256 \times 256$).

Our work provides a balanced and definitive characterization:

- **Birth-Death is a genuine accelerator in stationary Langevin dynamics**: on multimodal targets, BD achieves a $5\times$–$10\times$ reduction in required iterations to reach stationarity ($N \geq 64$), confirming the mathematical theory of Lu, Lu, and Nolen (2019). Furthermore, when diffusion score models suffer from severe systematic bias ($c = 0.5$), BD acts as a powerful target-aware corrector, cutting $\text{MMD}^2$ by $6.4\times$.
- **Repeated log-density computation is the central bottleneck for diffusion models**: because standard deep diffusion models only predict the score $\nabla \log p_t$, applying BD requires repeatedly evaluating an intractable marginal log-density at every sampling step.
- **Statistical and computational breakdown on non-toy data**: approximating $\log \pi_t$ via empirical reference sets or particle KDEs in high dimensions incurs prohibitive computational overhead ($O(N^2 D)$ or $O(NMD)$ per step) and collapses due to dimensional concentration (self-domination), inducing mode duplication and diversity loss (unique ratio dropping to 0.0625).
- **Resampling vs. Transport**: because BD is a spatial resampling operator rather than a temporal transport operator, it cannot advance particles across noise levels or reduce reverse inference steps (NFE).

By establishing the exact mathematical and computational boundary where interacting particle dynamics succeed (stationary sampling with known densities or biased score correction) versus where they fail (unconditional high-dimensional diffusion sampling with intractable densities), this study clarifies the fundamental distinction between transport and resampling in generative modeling.

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